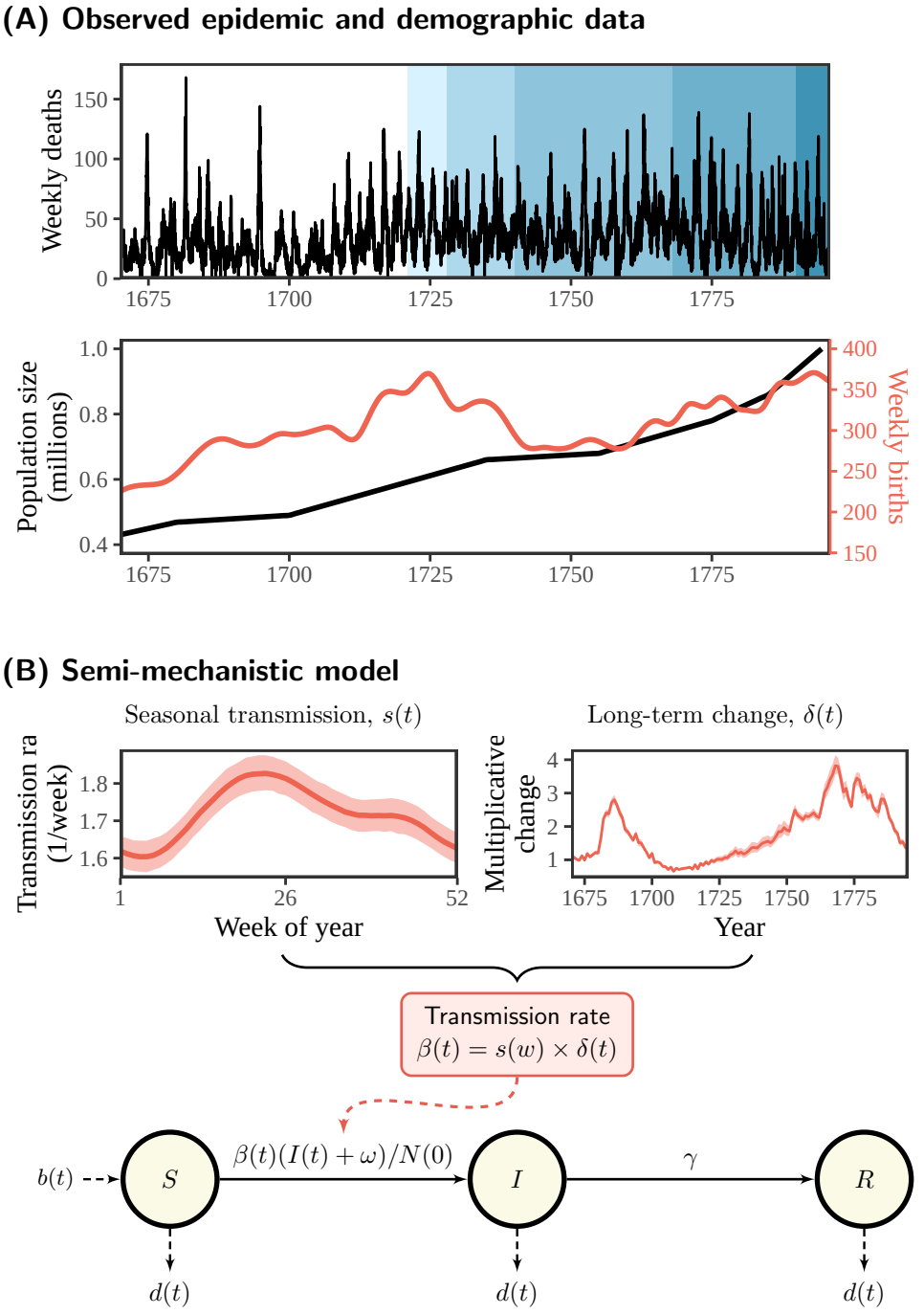

(A) Observed epidemic and demographic data
Weekly deaths
150
100
50
0
1675
1700
1725
1750
1775
Population size (millions)
1.0
0.8
0.6
0.4
Weekly births
400
350
300
250
200
150
(B) Semi-mechanistic model
Seasonal transmission, $s(t)$
Transmission ra (1/week)
1.8
1.7
1.6
1
26
52
Week of year
Long-term change, $\delta(t)$
Multiplicative change
4
3
2
1
Year
Transmission rate
$\beta(t) = s(w) \times \delta(t)$
$b(t)$
S
$\beta(t)(I(t)+\omega)/N(0)$
I
γ
R
$d(t)$
$d(t)$
$d(t)$